Instructions

1. Use this slide template as the pitch deck you will submit, please **do NOT** add any additional slides. Any additional slides may not be reviewed as part of your submission. You should have **6 slides total.**
2. The “Our Team” slide, slide 3, needs to be completely filled out.
3. Reminder- you must also submit:
    a. Your squad environment link
    b. Your demo / live app link
    c. Your Github repo link
    d. Recorded pitch video, under 3 minutes. **Please record yourselves briefly explaining/pitching your solution.** You can record this from any device.



Google Cloud

Agents for Impact

Hackathon



Table 10

Bronx Mobile Health Clinic Planner
Challenge 5 · Public Health & Resource Access
Bronx County, NY (FIPS 36005) · Google ADK + BigQuery

Our Team

Filming Availability: Should your team be selected as the winner, indicate if and when you are able to return to the Coca-Cola Roxy between 1:00-4:00 PM on Thursday 10/29 to film a team pitch of your project.
Project Sharing: Mark “yes” if you are OK with us sharing what you built with your team at work.
Share With: If “yes” to the above, is there anyone specific you’d like us to share it with at work? Provide email(s).
👉 **Check your Discord channel tonight to learn if you won. We will contact you no later than 9:00 PM.**

NAME	EMAIL	PHONE	FILMING AVAILABILITY (between 10AM-3PM on 10/29)	PROJECT SHARING	SHARE WITH (provide emails)
Michael Robards (Table 10)	mikero1999@gmail.com	[Add Phone #]	Yes (1:00–4:00 PM)	*Yes*	[Add Team Emails]

Challenge Overview

REAL-WORLD CHALLENGE: THE BRONX PARADOX

- 338 of 347 Bronx tracts are federal Shortage Areas (HPSA 16–19) despite 252 clinics within a 0.39 km (390m) median walk.
- Distance is NOT the barrier: 94% of tracts have 0 km uninsured distance penalty. Weekday 9–5 hours, 27.5% uninsured rates & 43.8% language barriers lock families out.
- Naive AI Traps: Standard spatial algorithms route vans to Rikers Island Jail (36005000100) or Bronx Psych Center (100% Group Quarters).

THE HUMAN STORY: DOÑA ELENA

- Age 58 · High Bridge (Tract 36005006700)
- Lives 400m from a clinic, works 6 days/wk as a home health aide, no car (86.2% of tract), Spanish-speaking.
- Cannot take an unpaid weekday off for 9–5 clinics—rations her asthma inhaler & skips blood pressure checks until a preventable $3,000 Saturday ER crisis.
- What she needs: Saturday walk-up clinic 2 blocks away at 165th & Ogden with Spanish CHWs and rapid point-of-care care.

WHO BENEFITS & SOCIETAL SCALE

- 1.47M Bronx Residents: Saturday walk-up chronic disease management, maternal/pediatric checks & rapid testing at subway hubs.
- Public Health Directors: Evidence-audited 4-Saturday deployment calendar across 7 Medicaid Exception HPSA zones.
- Frontline Health Workers: Automated 6th-grade bilingual (EN/ES) flyers, SMS text blasts & van intake scripts.
- National Blueprint: Replicable framework for every dense urban county where physical proximity ≠ actual care access.

Impact

OUR AGENT: 0%-DISTANCE NEED INDEX & DUAL-PERSONA UX

- 0%-Distance Need Composite: 30% Chronic Burden (asthma, diabetes, BP), 25% HPSA Severity (16–19), 25% Uninsured & Clinic Hours Gap, 20% Transit/Language/SVI.
- Hard Data Filters: Automatically rejects >25% Group Quarters & <500 pop to eliminate jail/psychiatric outliers (Rikers Island, Bronx Psych Center).
- Dual-Persona Synthesis: Director Briefing (rigorous 11-digit GEOID citations, scale caveats, equity ratios) vs. Community Outreach (strictly 6th-grade bilingual English/Spanish flyers, SMS alerts, CHW van intake scripts).
- Concrete Outcome for Doña Elena: Saturday #1 routes the van to 165th St & Ogden Ave (2 blocks from her home), staffing Spanish CHWs and stocking point-of-care rapid swabs and nebulizer kits.

VERIFIED 4-SATURDAY CLINIC SCHEDULE (19,977 POPULATION)

- Sat 1 (High Bridge · HPSA 18): Tract 36005006700 · 165th & Ogden (4/B/D, Bx13) · 27.5% Uninsured
- Sat 2 (Pelham · HPSA 19): Tract 36005008600 · White Plains Rd & Pelham Pkwy · 37.6% High BP
- Sat 3 (Fordham · HPSA 17): Tract 36005040302 · Grand Concourse & W 188th · 43.8% Ltd English
- Sat 4 (Hunts Point · HPSA 16): Tract 36005002300 · Hunts Pt Ave & Southern Blvd · 13.6% Asthma

POST-HOC DEMOGRAPHIC EQUITY AUDIT SCORECARD (PASS)

- Racial/Ethnic Representation: 93.9% Black + Hispanic (vs 84.8% Bronx baseline)
- Hispanic Representation: 69.8% (vs 56.6% Bronx baseline · 1.23x Equity Ratio ≥ 1.0x)
- Economic Need (Poverty): 38.6% (vs 27.4% Bronx baseline · 1.41x Equity Ratio ≥ 1.0x)
- Insurance Barrier: 24.3% Uninsured (vs 15.7% Bronx baseline · 1.55x Equity Ratio ≥ 1.0x)

Sources

THE BRONX BIGQUERY ECOSYSTEM (8 TABLES IN a4i_health)

• burden_tracts (347 tracts): CDC PLACES modeled prevalence (asthma, diabetes, hypertension, uninsured) + SVI vulnerabilities. Cleaned for -999 sentinels & group quarters.
• shortage_areas (338 tracts): HRSA HPSA designations across 7 Medicaid Exception zones (HPSA scores 16–19).
• shortage_facilities (8 FQHCs): Federal safety-net clinics.
• care_sites (252 sites): Clinic capacity and operating hours (8 to 112 hrs/week). Identifies weekend coverage deserts.
• exposure_tracts (361 tracts): EPA 12 km grid PM2.5 (mean 8.08–8.37 μg/m³; peak 57.0 μg/m³ for surge supplies).
• disease_weekly (212 weeks): NSSP syndromic surveillance (100% NULL in Bronx; mobile rapid testing sentinel).
• tract_access: Straight-line distance (median 390m, 94% have 0km gap; proves distance is flat in urban core).
• weather_forecast: Live Saturday meteorological & AQI posture.

THE 5 BRONX DATA AUDIT GUARDRAILS

① Urban Distance Trap (tract_access):
• Median clinic distance is 0.39 km (94% of tracts = 0 km uninsured penalty).
• Guardrail: Weighted at 0% in tract ranking; prioritizes HPSA severity, uninsured rates & operating hour gaps.

② Institutional / Group Quarters Trap (burden_tracts):
• Top raw disease burden tracts are Rikers Island Jail (36005000100) & Bronx Psych Center (100% Group Quarters).
• Guardrail: Hard-filters out tracts with >25% Group Quarters or <500 pop.

③ 12 km PM2.5 Grid Smoothing Trap (exposure_tracts):
• Annual mean_pm is flat (8.08–8.37 μg/m³, 1.33% CV) due to 12 km grid.
• Guardrail: Blocked from tract ranking; peak max_pm (57 μg/m³) + Live AQI configure mobile van nebulizer/spacer kits.

④ NSSP Dark-Signal Trap (disease_weekly):
• 100% NULL (0/212 weeks) for COVID/Flu/RSV in Bronx County data.
• Guardrail: Triggers Active Surveillance Blind-Spot Override; stocks point-of-care rapid swab kits on every van run.

⑤ Race Input Ban vs. Post-Hoc Equity Mandate:
• Guardrail: Planners banned from race inputs; context-isolated Auditor verifies ≥1.0x demographic parity post-hoc.

Prototype Architecture

GOOGLE AGENT DEVELOPMENT KIT (ADK) TOPOLOGY

7 Specialized Autonomous Agents Orchestrated on Vertex AI:

1. BurdenScout (Parallel): Analyzes chronic illness & SVI without racial features.
2. AccessScout (Parallel): Maps 7 ME HPSA zones & weekend operating hour deficits.
3. EnvSurveillanceScout (Parallel): Integrates peak PM2.5, weather & NSSP flags.
4. SchedulePlannerAgent (Loop): Synthesizes composite scores into 4-Saturday routes.
5. EquityAndScaleAuditorAgent (Loop): Isolated demographic context; checks 100% group quarters & post-hoc equity coverage.
6. DirectorBriefingAgent (Stage 3): High-precision policy briefing with GEOIDs & confidence intervals.
7. CommunityOutreachAgent (Stage 3): Warm, 6th-grade bilingual English/Spanish notices; bans technical acronyms.

STAGE 1: ParallelDataScouts (ADK ParallelAgent)
BigQuery (a4i_health) & Live APIs → Concurrent Domain Scouts
• BurdenScout: PLACES chronic burden + SVI (race banned) → state["burden_findings"]
• AccessScout: 252 care sites, 7 ME HPSA zones, MTA hubs → state["access_findings"]
• EnvSurveillanceScout: 12km PM2.5, NSSP blind spot, AQI → state["env_findings"]

STAGE 2: VerifiedScheduleLoop (ADK LoopAgent · max=3)
Planner ⇄ Context-Isolated Equity Auditor (include_contents="none")
• SchedulePlannerAgent: 4-Saturday calendar across distinct HPSA zones → state["proposed_schedule"]
• EquityAndScaleAuditorAgent: Context-isolated post-hoc equity audit (1.23x–1.55x) & scale rules. Emits escalate=True on PASS; re-plans on FAIL → state["audit_verdict"]

STAGE 3: Dual Contradictory Persona Synthesis
Specialized Outputs for Technical vs Community Stakeholders
• DirectorBriefingAgent: Rigorous evidence memo, 11-digit GEOIDs, scale caveats & equity audit → state["director_briefing"]
• CommunityOutreachAgent: 6th-grade bilingual (EN/ES) flyers, SMS text alerts, CHW scripts & Doña Elena narrative → state["community_notices"]

DELIVERY: 4-Tab Operations Center + Human-in-the-Loop Override
Production Gradio Web UI (FIPS 36005 · gemini-2.5-flash)
• Tab 1: Folium Interactive Map & 4-Saturday Calendar
• Tab 2: Director Briefing, 4-Resolution Audit & Data Traps
• Tab 3: Post-Hoc Demographic Equity Audit Scorecard
• Tab 4: 6th-Grade Bilingual Outreach Materials + 1-Click HITL Re-Planner